

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 1.3: Nutrition in Animals — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: Nutrition in Animals
Driving Question	DRIVING QUESTION: How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Ugali and Sukuma Wiki Puzzle — Launching the Anchoring Phenomenon	Learners observed a standard Kenyan school lunch (ugali, sukuma wiki, nyama choma) and noticed that three visibly different types of matter — starchy, fibrous, and protein-rich — enter the digestive system yet are all processed by the same body. They recorded specific anomalies: the body's apparent 'knowledge' of which digestive chemical to deploy, the different digestion times for starch versus protein, and the mystery of why the stomach does not digest itself.	Learners surfaced and organised their prior knowledge about digestion, distinguishing confident knowledge from uncertain hunches and genuine ignorance. They identified the major regions of the digestive system from memory, recognised that ugali is primarily starch, sukuma wiki is primarily fibre and vitamins, and nyama choma is primarily protein, and formulated at least one personal investigation question that they will pursue across the unit.	This lesson does not yet explain the phenomenon — it deliberately opens the puzzle. Learners' Draft 1 Food Journey Maps represent their current incomplete model of digestion. The three anomalies posed by the teacher have created cognitive dissonance that makes the rest of the unit's investigation necessary and motivating. The Driving Question Board now anchors all future learning to the central question: how is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail?
Lesson 2 Mechanical vs. Chemical Digestion — Breaking Down the Ugali and Sukuma Wiki	Learners observed (via animation) food being physically crushed and mixed by teeth, tongue, and stomach muscles, and (via the Mandazi Moment — chewing their own piece of mandazi for 60 seconds) that the taste of a starchy food changes from bland to slightly sweet, providing direct sensory evidence that a chemical change occurs in the mouth.	The body uses two distinct strategies to break down food: mechanical digestion (physical breakdown by teeth, tongue, and stomach churning — changes the size and shape of food but not its chemical identity) and chemical digestion (breakdown of food molecules by specific enzymes — salivary amylase in the mouth acts on starch, and pepsin with HCl in the stomach acts on protein). Each enzyme acts on a specific	This explains part of the ugali-sukuma wiki puzzle: the ugali (starch) begins to change chemically the moment it enters the mouth because salivary amylase is already present and specific to starch — which is why it tastes sweet after chewing. The nyama choma (protein) does not taste different in the mouth because amylase cannot act on protein. A new, deeper part of the puzzle is now exposed: if pepsin in the

		type of food molecule.	stomach dissolves protein, why doesn't the stomach dissolve its own protein-rich walls?
Lesson 3 Enzymes and Regions of the Alimentary Canal — The Body's Chemical Toolkit	Using Kenya Education Cloud resources and printed resource sheets, learners researched the specific enzymes active in each region of the alimentary canal and recorded their findings in region-by-region enzyme summary charts. They observed that different enzymes are produced by different glands/organs (salivary glands, gastric glands, pancreas, intestinal glands), that each enzyme acts on a specific substrate (starch, protein, or fat), and that the number and variety of enzymes increases dramatically from mouth to small intestine.	The body deploys a precise, region-specific chemical toolkit to digest different food types: salivary amylase begins starch digestion in the mouth; pepsin and rennin break down proteins in the acidic stomach environment; and the small intestine receives pancreatic amylase, lipase, and trypsin from the pancreas, plus peptidases from the intestinal lining itself, completing the digestion of all three macronutrients. Each enzyme is specific to its substrate — salivary amylase cannot digest protein, and pepsin cannot digest starch. This enzyme specificity is a key adaptation of the digestive system.	This explains the first major puzzle from the anchoring phenomenon: the body does NOT release one general-purpose chemical into the digestive system. Instead, it 'knows' which chemical to deploy at each stage because specific enzymes are produced by specific glands and secreted only at specific locations along the alimentary canal. The ugali's starch, the nyama choma's protein, and the fat in the cooking oil are each handled by different enzymes in different regions — this regional chemical specialisation is the body's solution to the challenge of digesting a complex, mixed Kenyan meal.
Lesson 4 Food Tests Practical Part 1 — Detecting Starch and Reducing Sugars in Kenyan Foods	Learners observed that ugali (both cooked and raw maize flour) turned blue-black with iodine solution, confirming the presence of starch; that ripe banana and glucose solution turned brick-red with Benedict reagent, confirming reducing sugars; and that sukuma wiki extract gave weak or negative results for both tests, which raised a new question about what nutrients sukuma wiki actually provides.	Learners learned that the iodine test and Benedict test are reliable, reproducible chemical tests for specific carbohydrate types; that starch and reducing sugars are chemically different and require different enzymes (amylase for starch, other enzymes for resulting sugars); and that the ripening of a banana involves conversion of starch to reducing sugars, linking biology to the chemistry of everyday Kenyan foods.	Learners explained that the starch in ugali is the primary substrate for salivary amylase in the mouth and pancreatic amylase in the small intestine; that the reducing sugars produced by amylase action represent the product of successful carbohydrate digestion; and that their food test evidence directly supports the enzyme-specificity principle established in Lesson 3 — each enzyme acts on a specific substrate, and the substrates are chemically real and detectable in the actual foods of the anchoring phenomenon meal.
Lesson 5 Food Tests Practical (Part 2) — Detecting Proteins and Fats in the Kenyan Meal	Learners observed that nyama choma extract and milk turned violet with Biuret reagent (positive for protein), while ugali extract remained blue (negative). Butter and cooking oil produced a milky-white emulsion with the ethanol emulsion test (positive for fat), while sukuma wiki gave a faint emulsion and ugali gave none. These results were consistent across all class groups.	Proteins contain peptide bonds that coordinate with copper ions in Biuret reagent to produce a violet colour. Fats dissolve in ethanol and form a milky emulsion when the ethanol solution is poured into water. The presence of protein in nyama choma explains why the stomach secretes pepsin (a protease) activated by hydrochloric acid. The presence of fat explains why bile salts (emulsification) and lipase (chemical digestion) are both needed in the small intestine — fat digestion is delayed compared to starch and protein digestion because fat must be emulsified first.	The Ugali and Sukuma Wiki meal contains all four major nutrient classes: starch (confirmed in Lesson 4), reducing sugars (Lesson 4), protein (today's Biuret test on nyama choma and bean broth), and fat (today's ethanol emulsion test on cooking oil and butter). This explains part of the puzzle: the body 'knows' which chemical to release at each stage because each region of the alimentary canal is matched to the specific nutrient chemistry present — salivary amylase in the mouth for starch, pepsin in the stomach for protein, and bile plus lipase in the small intestine for fat. The meal's nutrient profile drives the digestive

			system's regional specialisation.
Lesson 6 Adaptations of the Human Digestive System — Structure Meets Function	Learners examined labelled diagrams and photomicrographs of three structural features: (1) villi and microvilli (brush border) of the small intestinal wall, showing finger-like projections, internal capillary networks, and lacteals; (2) rugae folds of the stomach wall and the goblet-cell mucus layer; and (3) the cardiac and pyloric sphincters at the entry and exit of the stomach.	The small intestine's villi and microvilli increase its absorptive surface area by up to 600-fold, allowing a 3 cm wide tube to absorb all glucose (from ugali starch), amino acids, and fatty acids (from nyama choma) efficiently. The stomach's rugae allow it to expand and maximise contact with food during churning, while the mucus lining (from goblet cells) and the secretion of pepsin as inactive pepsinogen prevent the stomach from digesting its own wall. The cardiac sphincter prevents stomach acid from refluxing into the oesophagus, while the pyloric sphincter controls the rate at which partially digested chyme enters the small intestine, ensuring enzymes have time to act.	The puzzle of how a narrow small intestine absorbs all nutrients from a Kenyan meal is resolved by structural adaptation: massive surface area amplification through villi and microvilli means the intestinal lining contacts and absorbs digested nutrients at an extraordinary rate. The stomach's self-protection mechanisms (mucus, pepsinogen) explain why the body does not digest itself despite producing powerful acid and proteases. The sphincters explain how the body 'knows' when to move food from one region to the next — they are mechanical gatekeepers that coordinate timing across the alimentary canal. Together, these structural adaptations answer two of the original DQB questions from Lesson 1 and set up the investigation of what happens when these adaptations fail.
Lesson 7 Model Building — Making a 3D Model of the Human Digestive System	Learners observed — through the act of physical construction — that the alimentary canal is a continuous, regionally specialised tube in which each section has a distinct shape, size, and lining (e.g., thick muscular stomach wall vs. thin, villi-lined small intestine) that directly corresponds to its function. They also observed, via peer-assessment feedback, which organs and processes their classmates struggled most to represent accurately (commonly: accessory organs — liver, gall bladder, pancreas — and the dual role of bile as an emulsifier, not an enzyme).	Learners consolidated and integrated knowledge from six prior lessons into a single explanatory model: (1) digestion is both mechanical (teeth, stomach churning, peristalsis) and chemical (enzyme-catalysed) and both strategies are needed to fully process ugali, sukuma wiki, and nyama choma; (2) enzymes are region-specific — salivary amylase acts in the mouth on the starch in ugali, pepsin acts in the stomach's acidic environment on the proteins in nyama choma, and pancreatic lipase (after bile emulsification) acts in the small intestine on fats; (3) the small intestine's villi are the structural adaptation that solves the 'invisible machinery' puzzle — massively increasing surface area for absorption without increasing the tube's outer diameter; and (4) sphincters act as valves that control the rate of movement between regions, giving each section time to complete its specific task.	The 3D model provides a unified physical answer to the anchoring phenomenon: the ugali-and-sukuma-wiki meal does not encounter 'single uniform processing' — instead, it passes through a precisely sequenced series of chemically distinct micro-environments, each releasing the right enzyme at the right pH for the right substrate. The body 'knows' which chemical to release at each stage because each organ's cells are genetically programmed to secrete specific enzymes in response to specific food stimuli (a concept flagged as extending beyond this lesson). Carbohydrates (ugali starch) are partially digested in the mouth and completed in the small intestine; proteins (nyama choma) require the stomach's acid activation of pepsinogen before digestion can begin — explaining why protein digestion takes longer. The villi of the small intestine are the answer to how a 3 cm tube absorbs all nutrients without 'visible machinery': the machinery is microscopic, folded into the wall itself.

Lesson 8 DQB Synthesis & Assessment — Driving Question Resolution and Unit Wrap-Up	Learners observed a brick-red/orange colour change when Benedict's solution was added to a simulated digested-ugali sample (glucose solution) and heated in a water bath — providing direct chemical evidence that the starch in ugali has been broken down into reducing sugars (glucose) by the action of amylase enzymes during digestion.	The human digestive system is structurally and chemically adapted at every region to process the specific nutrients in a Kenyan meal: mechanical digestion (teeth, stomach churning) and chemical digestion (region-specific enzymes — salivary amylase, pepsin, pancreatic amylase, lipase, trypsin) break ugali starch, sukuma wiki fibre, and nyama choma protein into absorbable monomers; villi and microvilli of the ileum (surface area increased up to 600-fold) absorb these monomers into the bloodstream; and protective adaptations (mucus lining, HCl regulation, sphincters) prevent self-digestion and control the rate of passage.	The Ugali and Sukuma Wiki Puzzle is now fully explained: (1) The body 'knows' which chemical to release at each stage because digestion is controlled by a precise, region-specific enzyme system — each enzyme is secreted only where its substrate arrives, triggered by hormonal and neural signals; (2) Carbohydrates digest faster than proteins because salivary and pancreatic amylase act on the relatively simple glycosidic bonds of ugali starch from the moment the meal enters the mouth, while proteins in nyama choma require the more complex, sequential action of pepsin (acidic, stomach), trypsin and chymotrypsin (alkaline, small intestine); (3) The body avoids digesting itself through mucus lining the stomach and intestinal walls, secretion of inactive enzyme precursors (zymogens) that activate only in the correct environment, and the alkaline mucus secreted by Brunner's glands in the duodenum that neutralises stomach acid before it contacts the intestinal wall.
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ATTRIBUTION AND LICENSING

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